SP19-2: Poincare Paper Outline

Working Title: "The Wallach Bottleneck and the Poincare Conjecture: Why S^3 Is Forced" Lead: Lyra + Elie Target: Geometric and Functional Analysis (GAFA) Date: May 13, 2026

Scope (honest)

This paper does NOT claim an independent proof of Poincare. It claims: 1. BST provides the structural explanation for WHY Perelman's proof works 2. Every numerical quantity in the Poincare proof chain is a BST integer or ratio 3. For compact simply-connected M^3 embedded in Q^5, the BST argument is complete without Ricci flow 4. For the general case, Perelman's Ricci flow is still required, but BST explains why it converges

Sections (10)

1. Introduction and Main Result

Thesis: D_IV^5 structural data explains Poincare. Wallach bottleneck forces S^3. Not alternative proof — structural explanation. - Ref: GC-2 template mapping

2. The BST Integers in the Poincare Landscape

18+ appearances (conservative count):

#	Quantity	Value	BST
1	Manifold dimension	3	N_c
2	Thurston geometry count	8	2^N_c
3	Excluded geometries	7	g
4	Exclusion conditions	5	n_C
5	Ambient real dimension Q^5	10	2*n_C
6	Codimension M^3 in Q^5	7	g
7	Whitney embedding dim	7	2*N_c+1
8	R(S^3)	6	C_2
9	Bergman spectral gap	6	C_2
10	Lichnerowicz lambda_1	3	N_c
11	Perelman entropy constant	3/2	N_c/rank
12	Ricci decay rate	6	C_2
13	Wallach parameters	6	C_2
14	Sym^2(T*M) dimension	6	C_2
15	Total II components	42	C_2*g
16	Smale easy threshold	5	n_C
17	Freedman intermediate dim	4	rank^2
18	K41 branching ratio	5/3	n_C/N_c

Plus: normal bundle (2,3,2) = (n_C-N_c, N_c, n_C-N_c), sum = g. Stability eigenvalues {1, N_c, 1}. - Toys: 2138 (16/16), 2143 (11/11), 2145 (17/17). Tier: D.

3. Thurston Exclusions as Wallach Kernel

8 = 2^N_c geometries. Image = 1-dim (S^3). Kernel = g = 7. Each excluded geometry fails one of n_C = 5 conditions. - Toy 2143 (11/11), T1818. Tier: I. - Gap: Wallach kernel identification is structural, not representation-theoretic.

4. The Wallach Bottleneck Theorem (T1829)

Three selection equations uniquely force n=5. At selected dimension, pi_2 organizes S^3 spectral theory. - Toy 2151 (26/26), T1829. Tier: D.

5. Ricci Flow as Wallach Spectral Evolution

Cumulative identity: sum_{j=0}^m (j+1)^2 = dim H_m(R^5). K41 = n_C/N_c = 5/3. Decay rate = C_2 = 6. - Toy 2145 (17/17), T1823. Tier: I-to-C. - Gap: Spectral organization implies convergence (not yet proved).

6. The Berger-Klingenberg Route (Approach B)

Q^5 sectional curvatures in [1, rank^2] = [1, 4]. Pinching delta = 1/4 at Berger boundary. Positive II helps. C_2 parameters = C_2 constraints (square system). - Toy 2148 (16/16). Tier: C. - Gap: Rigorous II positivity at Wallach level.

7. Stability of the Totally Geodesic Embedding (FC-3a)

TG S^3 in Q^5 is strictly stable. Normal bundle = g = 7. All stability eigenvalues positive. Unique minimal M^3 at Wallach level. - Toys 2152 (20/20), 2153 (13/13). Tier: D for Q^5 submanifolds. - Gap: General M^3 must admit Wallach-level embedding in Q^5.

8. The BST-Perelman Hybrid Proof

Synthesis: Perelman provides dynamics (Ricci flow), BST provides structure (Wallach forces endpoint). Five-step argument combining both. - All toys. Tier: C for general case, D for Q^5 case.

9. Over-Determination and the Confinement Parallel

5 monotonicity controls for 1 outcome = 5:1. d = N_c = 3 is both Poincare and confinement dimension. R(S^3) = C_2 is both scalar curvature and Casimir eigenvalue. - Tier: I (parallel structural, not proved).

10. Honest Scope and Open Questions

- PROVED: 18+ BST integers, T1829, stability, cumulative identity
- CONDITIONAL: Berger-Klingenberg, spectral embedding, Ricci-Wallach connection
- REQUIRES PERELMAN: General case
- OPEN: Bergman flow on Q^5 -> Ricci flow? (genuinely new math)

Toy/Theorem Map

Section	Toys	Tests	Tier
2	2138, 2143	27/27	D/I
3	2143	11/11	I
4	2151	26/26	D
5	2145	17/17	I-C
6	2148	16/16	C
7	2152, 2153	33/33	D/C
Total	7 toys	130/130	—

Gaps for New Work

1. Gauss-Codazzi determinant (Sections 6-7): Weyl tensor of Q^5 on 3-planes. Tractable linear algebra. Would upgrade C->D.
2. Spectral embedding existence (Section 7): Every simply-connected M^3 admits Wallach-level embedding in Q^5. Berard-Besson-Gallot candidate.
3. Wallach kernel formalization (Section 3): Representation-theoretic kernel computation. Upgrade I->C/D.
4. Ricci-Wallach convergence (Section 5): K-type filtration implies PDE convergence. New mathematics.
5. Bergman flow = Ricci flow (Section 10): Most ambitious. If proved: fully BST-native Poincare.

Venue

GAFA (Geometric and Functional Analysis) — intersection of geometric analysis and representation theory. Does not claim to replace Perelman (would need Annals), but explains him structurally.